



# 中国科学技术大学

数统方程 UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

第二周作业

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6. 解:  $u = u(x, y, z)$

$$(1) \frac{\partial u}{\partial y} + a(x, y)u = 0 \Rightarrow \ln|u| = -\int a(x, y)dy + C_1(x, z) \quad (\text{关于 } y \text{ 积分})$$

$$\Rightarrow u(x, y, z) = C_2(x, z)e^{-\int a(x, y)dy} \quad (\text{拆去 } |u| \text{ 绝对值后, } C_2(x, z) \text{ 不一定恒正})$$

$$(2) u_{xy} + u_y = 0 \xrightarrow{v=u_y} v_x + v = 0$$

$$\Rightarrow v = C_1(y, z)e^{-x}$$

$$u_y = C_1(y, z)e^{-x} \Rightarrow u(x, y, z) = e^{-x} \int C_1(y, z)dy + C_2(x, z)$$

$$\text{即解为 } u(x, y, z) = C_3(y, z)e^{-x} + C_4(x, z)$$

$$(3) u_{tt} = a^2 u_{xx} + 5x^2 \quad \begin{cases} \text{一个形如 } v(x) \text{ 的特解} \\ \text{齐次方程 } u_{tt} = a^2 u_{xx} \text{ 通解 } w(x, t) \end{cases} \Rightarrow u(x, t) = v(x) + w(x, t)$$

$$v(x) \text{ 满足: } 0 = v_{tt} = a^2 v_{xx} + 5x^2, \text{ 取 } v(x) = -\frac{x^4}{4a^2}$$

$$w(x, t) \text{ 满足: } w_{tt} = a^2 w_{xx} \Rightarrow w(x, t) = f(x+at) + g(x-at)$$

$$\text{故 } u(x, t) = f(x+at) + g(x-at) - \frac{x^4}{4a^2}$$

$$7. \text{解: } (1) \begin{cases} u_t = a^2 u_{xx} & 0 < x < l, t > 0 \quad (\text{热方程}) \\ u_x(t, 0) = 0 & t > 0 \quad (\text{绝热}) \\ u(t, l) = u_0 & t > 0 \quad (\text{保持常温 } u_0) \\ u(0, x) = \varphi(x) & 0 < x < l \quad (\text{初始温度}) \end{cases}$$

$$(2) \begin{cases} u_t = a^2 u_{xx} & 0 < x < l, t > 0 \\ u(0, x) = \varphi(x) & 0 < x < l \\ -ku_x(t, 0) = q_1; \quad ku_x(t, l) = q_2 & t > 0 \end{cases} \quad \begin{aligned} &\text{热传导定律: } \frac{\partial u}{\partial n} = -\frac{q}{k} \text{ 中} \\ &\vec{n} \text{ 外法向 } (-u_x(t, 0); u_x(t, l)) \\ &q: \text{ 逆界向外流出的密度} \\ &(\text{本题中为 } -q_1, -q_2) \end{aligned}$$

$$(3) \begin{cases} u_t = a^2 u_{xx} & 0 < x < l, t > 0 \\ u(0, x) = \varphi(x) & 0 < x < l \\ u(t, 0) = \mu(t) & t > 0 \\ ku_x(t, l) + h(u(t, l) - u_0) = h(u_0 - u_0) & t > 0 \end{cases} \quad \begin{aligned} &\frac{\partial u}{\partial n} = -\frac{q}{k} \quad \text{其中 } \frac{\partial u}{\partial n}|_{x=l} = u_x|_{x=l} \\ &q = h(u - u_0) \end{aligned}$$





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8. 解:

$$\begin{cases} u_{tt} = a^2 u_{xx} & t > 0, 0 < x < l & \text{(波动方程)} \\ u(t, 0) = u(t, l) = 0 & t > 0 & \text{(两端为固定)} \\ u_t(0, x) = 0 & 0 < x < l & \text{(初始速度为0)} \\ u(0, x) = \frac{2h}{l}x & 0 < x < \frac{l}{2} \\ u(0, x) = \frac{2h}{l}(l-x) & \frac{l}{2} < x < l & \text{("近似为三角形")} \end{cases}$$

9. 解: (1)  $u(t, x) = x^2(t+1)$

$$(2) \begin{cases} u_{tt} = a^2 \Delta_x u \\ u|_{t=0} = \varphi(x) \\ u_t|_{t=0} = \psi(x) \end{cases} \Rightarrow \begin{cases} u_{tt} = a^2(u_{rr} + \frac{2}{r}u_r) \\ u(0, r) = \varphi(r) \\ u_t(0, r) = \psi(r) \end{cases}$$

令  $V = ru$ , 有  $\begin{cases} V_{tt} = a^2 V_{rr} \\ V(0, r) = r\varphi(r) \\ V_t(0, r) = r\psi(r) \end{cases}$  达朗贝尔公式  $\Rightarrow V = \frac{(r+at)\varphi(r+at) + (r-at)\varphi(r-at)}{2} + \frac{1}{2a} \int_{r-at}^{r+at} s\psi(s)ds$

$$u(t, r) = \frac{(r+at)\varphi(r+at) + (r-at)\varphi(r-at)}{2r} + \frac{1}{2ar} \int_{r-at}^{r+at} s\psi(s)ds$$

(3) 考虑  $u = [(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{-\frac{1}{2}}$  其满足  $\Delta_x u = 0$

$$x^2 + y^2 + z^2 = 1 \text{ 时 } u = (1+y)^{-\frac{1}{2}} \text{ 得 } \begin{cases} x_0 = 0 \\ y_0 = -2 \\ z_0 = 0 \end{cases}$$

$$\text{故 } u = [x^2 + (y+2)^2 + z^2]^{-\frac{1}{2}}$$

(唯一性由拉普拉斯方程 Dirichlet 边值问题解的存在唯一性保证)



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10. 解: 可设  $u(t, x) = v(t, x) + w(t, x)$  (叠加原理)

$$\text{其中 } v(t, x): \begin{cases} \frac{\partial v}{\partial t} + a \frac{\partial v}{\partial x} = 0 \\ v(0, x) = \varphi(x) \end{cases}$$

$$w(t, x): \begin{cases} \frac{\partial w}{\partial t} + a \frac{\partial w}{\partial x} = f(t, x) \\ w(0, x) = 0 \end{cases}$$

$$\text{令 } \begin{cases} \xi = x - at \\ \eta = t \end{cases} \Rightarrow \begin{cases} \frac{\partial v}{\partial t} = \frac{\partial v}{\partial \eta} - a \frac{\partial v}{\partial \xi} \\ \frac{\partial v}{\partial x} = \frac{\partial v}{\partial \xi} \end{cases}$$

$$\text{由齐次化原理, } w(t, x) = \int_0^t f(\tau, x - a(t - \tau)) d\tau$$

$$\text{注: } w(t, x) = \int_0^t g(t, x; \tau) d\tau$$

代入有  $v_\eta = 0$ .

从而  $v(\xi, \eta) = F(\xi)$

换回  $x, t$  有  $v(t, x) = F(x - at)$

结合  $v(0, x) = \varphi(x)$  知  $F(x - at) = \varphi(x - at)$

故  $v(t, x) = \varphi(x - at)$

其中  $g(t, x; \tau)$  为齐次方程:

$$\begin{cases} \frac{\partial g}{\partial t} + a \frac{\partial g}{\partial x} = 0 \\ g(t, x; \tau) \big|_{t=\tau} = f(\tau, x) \end{cases} \text{的解}$$

$$\Rightarrow g(t, x; \tau) = f(\tau, x - a(t - \tau))$$

故  $u(t, x) = v(t, x) + w(t, x)$

$$= \varphi(x - at) + \int_0^t f(\tau, x - a(t - \tau)) d\tau$$



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